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Ramos

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(54) **SUPPORT APPARATUS FOR A CURTAIN ROD**

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B25H 1/00 (2006.01)

(52) **U.S. Cl.**
CPC ***A47H 7/00*** (2013.01); ***B25H 1/0035*** (2013.01); ***B25H 1/0042*** (2013.01)

(58) **Field of Classification Search**
CPC A47H 7/00; B25H 1/0035; B25H 1/0042
See application file for complete search history.

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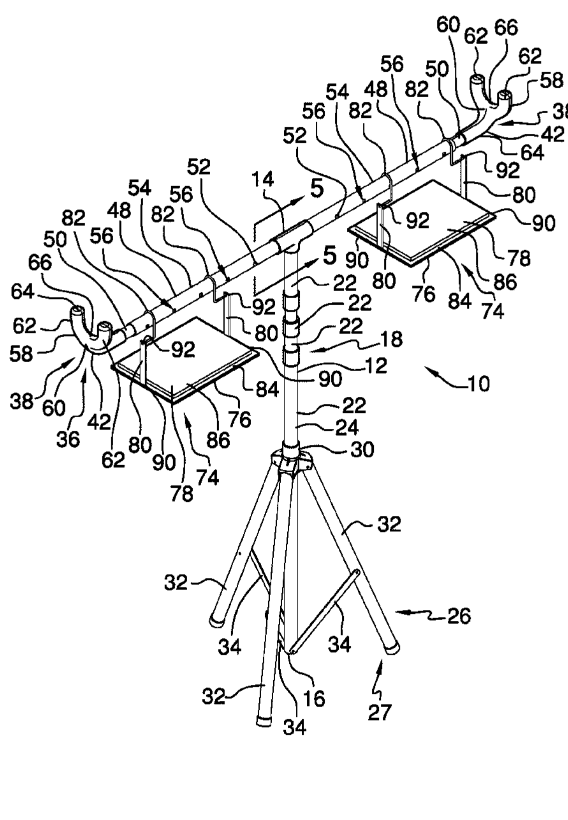
Primary Examiner — Lee D Wilson

(57) **ABSTRACT**

A support apparatus for supporting a curtain rod in a horizontal orientation during installation of the curtain rod includes a post with a top end and a bottom end which is extendable and retractable. Each one of a pair of extension arms is coupled to the post adjacent to the top end and extends outwardly from the post in one of a pair of opposing directions. Each one of a pair of saddles is coupled to a distal end of an associated extension arm with respect to the post. Each saddle comprises a central member that is coupled to and extends upwardly from the associated extension arm. An upper surface of the central member extends transversely with respect to the extension arm and is configured for supporting the bar, with the pair of saddles being configured for jointly supporting the bar in a horizontal orientation.

18 Claims, 5 Drawing Sheets



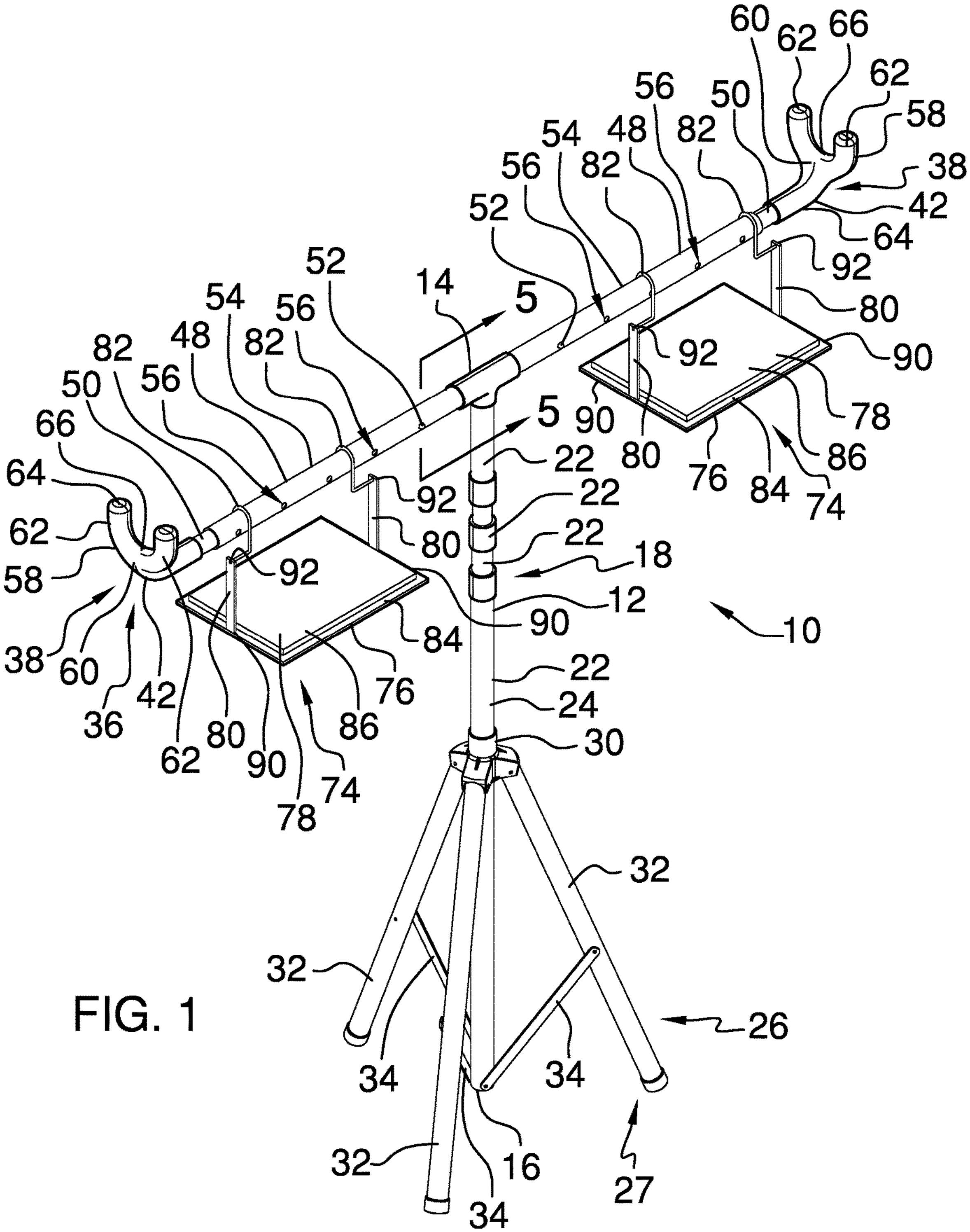

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FIG. 1

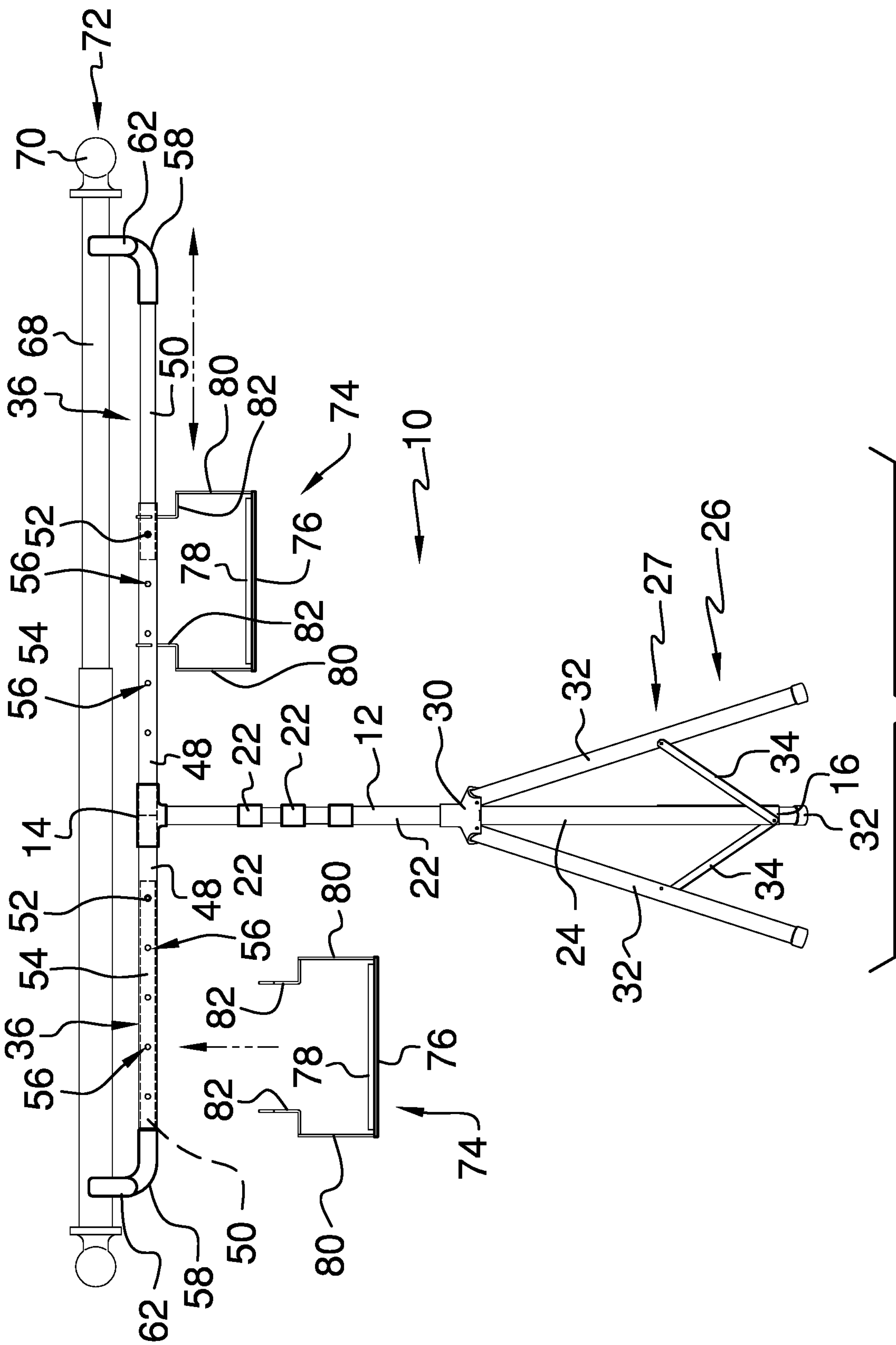

FIG. 2

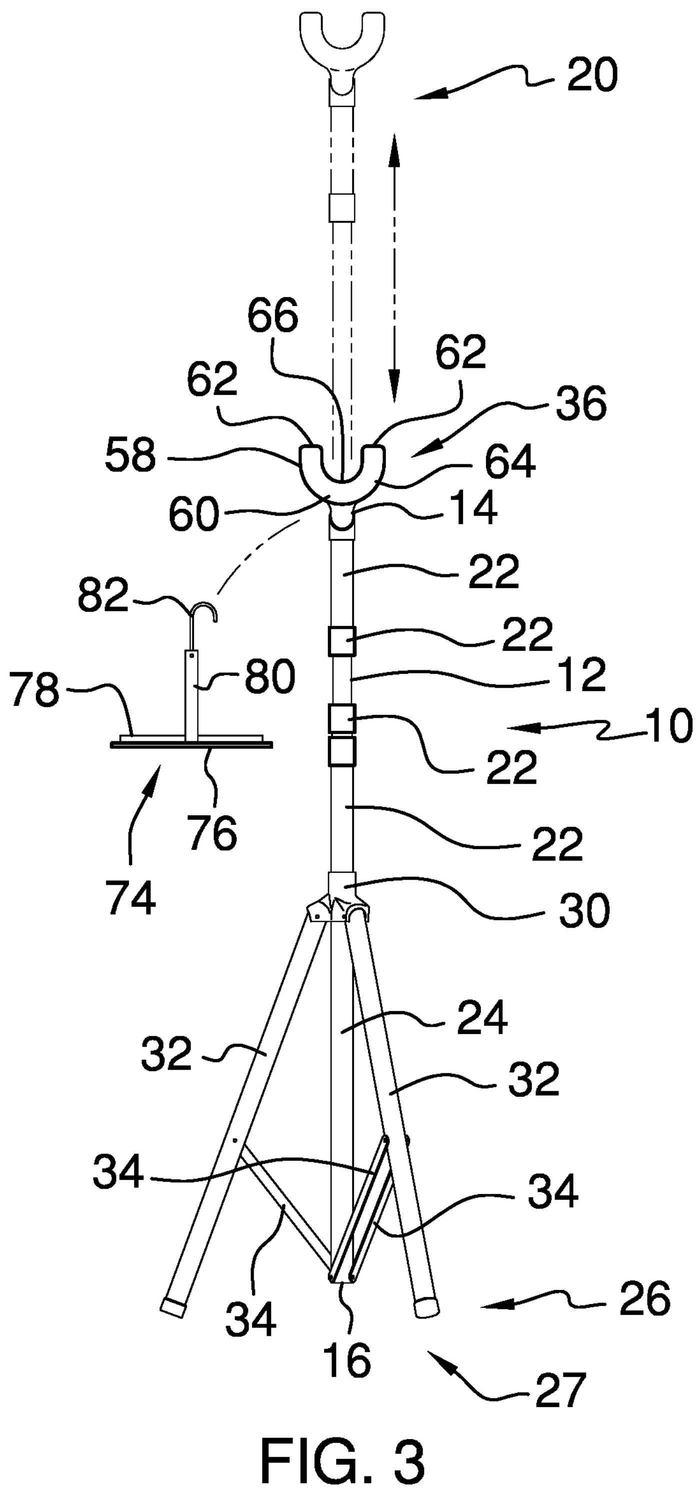
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FIG. 3

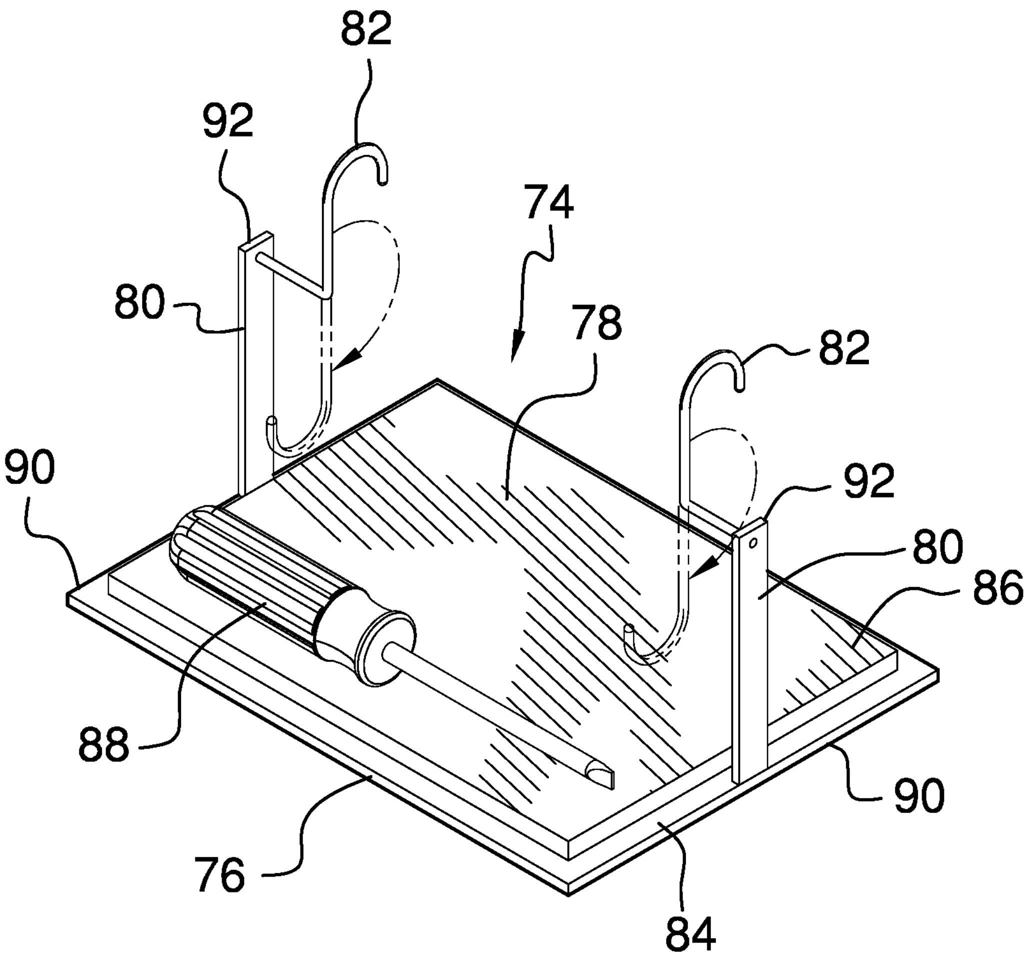

FIG. 4

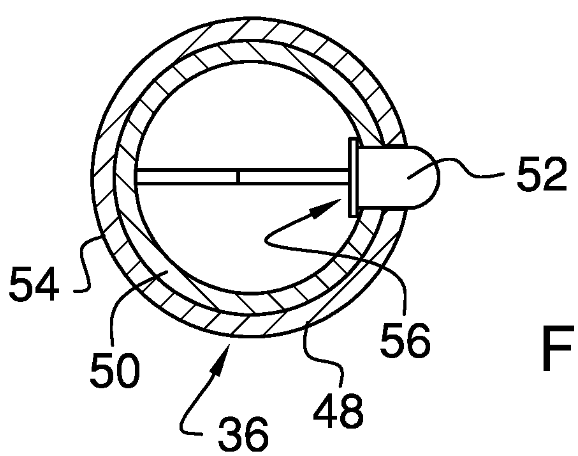

FIG. 5

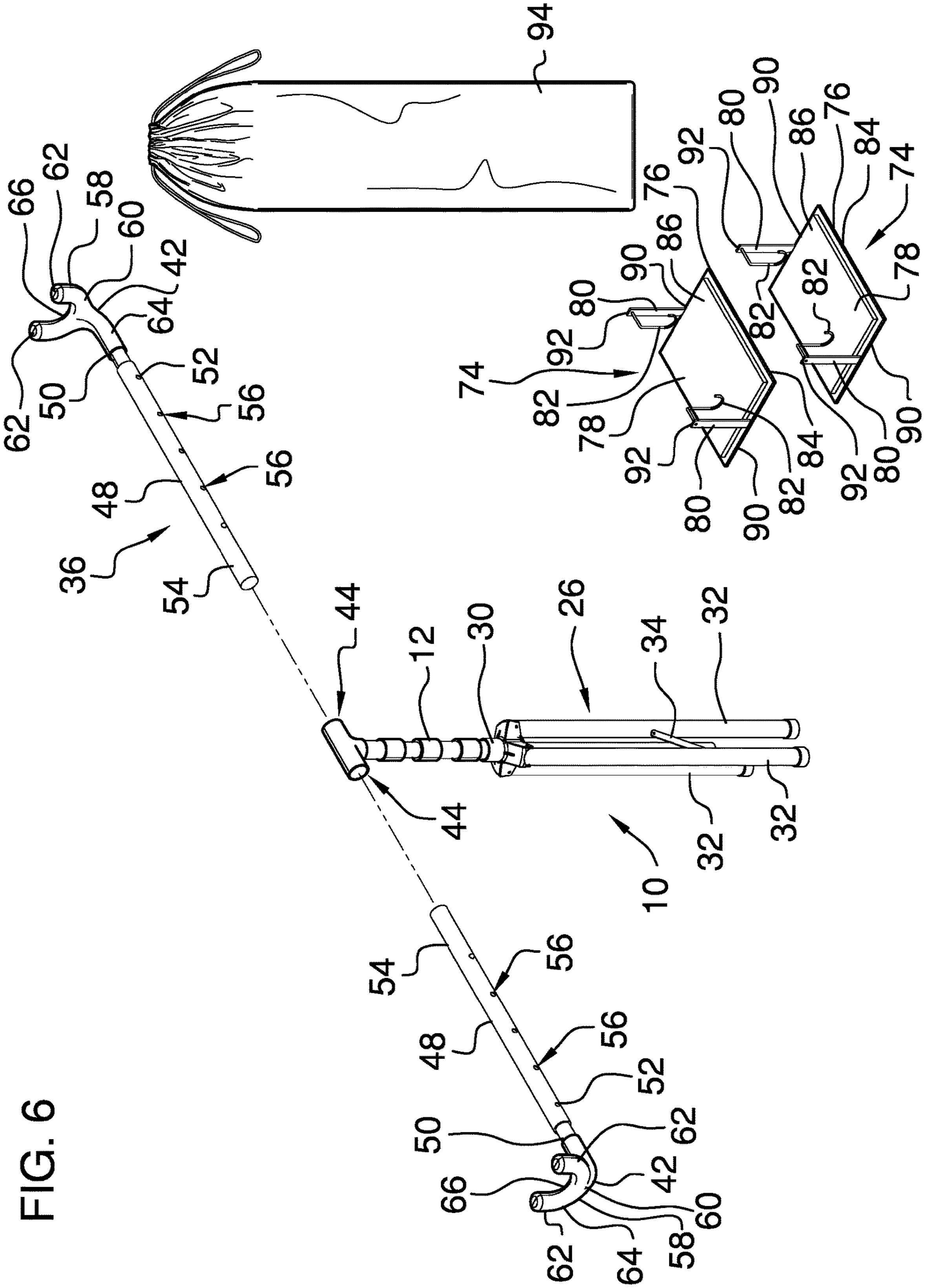

FIG. 6

SUPPORT APPARATUS FOR A CURTAIN ROD

CROSS-REFERENCE TO RELATED APPLICATIONS

Not Applicable

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

Not Applicable

THE NAMES OF THE PARTIES TO A JOINT RESEARCH AGREEMENT

Not Applicable

INCORPORATION-BY-REFERENCE OF MATERIAL SUBMITTED ON A COMPACT DISC OR AS A TEXT FILE VIA THE OFFICE ELECTRONIC FILING SYSTEM

Not Applicable

STATEMENT REGARDING PRIOR DISCLOSURES BY THE INVENTOR OR JOINT INVENTOR

Not Applicable

BACKGROUND OF THE INVENTION

(1) Field of the Invention

The disclosure relates to support apparatuses and more particularly pertains to a new support apparatus for supporting a curtain rod in a horizontal orientation during installation of the curtain rod.

(2) Description of Related Art Including Information Disclosed Under 37 CFR 1.97 and 1.98

The prior art relates to support apparatuses including one device disclosed by United Kingdom patent GB 13,843 which comprises a telescopic post mounted on a fixed tripod with a pair of pivotally mounted arms which extend outwardly from the post for supporting a curtain rod. As disclosed in the patent, the device is intended to be lifted by a user to place the curtain rod on mounting hooks above a window.

BRIEF SUMMARY OF THE INVENTION

An embodiment of the disclosure meets the needs presented above by generally comprising a post with a top end and a bottom end which is extendable between a retracted position and an extended position. Each one of a pair of extension arms is coupled to the post adjacent to the top end and extends outwardly from the post in one of a pair of opposing directions. Each one of a pair of saddles is coupled to a distal end of an associated extension arm with respect to the post. Each saddle comprises a central member that is coupled to and extends upwardly from the associated extension arm. An upper surface of the central member extends transversely with respect to the extension arm and is configured for supporting the bar, with the pair of saddles being configured for jointly supporting the bar in a horizontal orientation.

There has thus been outlined, rather broadly, the more important features of the disclosure in order that the detailed description thereof that follows may be better understood, and in order that the present contribution to the art may be better appreciated. There are additional features of the disclosure that will be described hereinafter and which will form the subject matter of the claims appended hereto.

The objects of the disclosure, along with the various features of novelty which characterize the disclosure, are pointed out with particularity in the claims annexed to and forming a part of this disclosure.

BRIEF DESCRIPTION OF SEVERAL VIEWS OF THE DRAWING(S)

The disclosure will be better understood and objects other than those set forth above will become apparent when consideration is given to the following detailed description thereof. Such description makes reference to the annexed drawings wherein:

FIG. **1** is a top front side perspective view of a support apparatus according to an embodiment of the disclosure.

FIG. **2** is a front in-use view of an embodiment of the disclosure.

FIG. **3** is a side view of an embodiment of the disclosure.

FIG. **4** is a detail in-use view of a tool tray according to an embodiment of the disclosure.

FIG. **5** is a detail cross-sectional view of an extension arm according to an embodiment of the disclosure from the direction of Arrows **5-5** in FIG. **1**.

FIG. **6** is an exploded view of an embodiment of the disclosure.

DETAILED DESCRIPTION OF THE INVENTION

With reference now to the drawings, and in particular to FIGS. **1** through **6** thereof, a new support apparatus embodying the principles and concepts of an embodiment of the disclosure and generally designated by the reference numeral **10** will be described.

As best illustrated in FIGS. **1** through **6**, the support apparatus **10** generally comprises a post **12** has a top end **14** and a bottom end **16**, and the post **12** is extendable between a retracted position **18** and an extended position **20**. The post **12** has a height between the top end **14** and the bottom end **16** that is less than 3.5 feet when the post **12** is positioned in the retracted position **18**, and the height is greater than 7.5 feet when the post **12** is positioned in the extended position **20**. The post **12** comprises a plurality of rods **22**. Each rod **22** except for a bottommost rod **24** of the plurality of rods **22** is telescopically received into an adjacently lower rod **22** of the plurality of rods **22**. Each rod **22** except for the bottommost rod **24** is frictionally retained by the adjacently lower rod **22** with respect to the adjacently lower rod **22**, but the rods **22** may alternatively be secured by a clamp, set screw, spring-biased pin, or the like.

A collapsible base **26** is coupled to the bottom end **16** of the post **12** for supporting the post **12** on a support surface **28**. The collapsible base **26** is positioned in a deployed position **27** such that the collapsible base **26** extends outwardly from the post **12**. The collapsible base **26** is collapsible toward the post **12**. The collapsible base **26** comprises a slider **30**, a plurality of legs **32**, and a plurality of links **34**.

The slider **30** is slidably coupled to and surrounds the post **12**. Each leg **32** is pivotally coupled to the slider **30** such that each leg **32** is pivotable toward and away from the post **12**. Each leg **32** extends outwardly from the post **12** and is angled downwardly. Each link **34** is pivotally coupled to the post **12** adjacent to the bottom end **16** and to one of the plurality of legs **32** such that each leg **32** pivots outwardly away from the post **12** when the slider **30** moves toward the bottom end **16** and toward the post **12** when the slider **30** moves away from the bottom end **16**.

Each one of a pair of extension arms **36** is coupled to the post **12** adjacent to the top end **14**. Each extension arm **36** extends outwardly away from the post **12** in one of a pair of opposing directions. The extension arms **36** are extendable away from and retractable toward the post **12** and are positionable in a shortened position **38** and a lengthened position **40**. When the arms are in the shortened position **38**, an arm span distance between a distal end **42** of one extension arm **36** with respect to the post **12** and a distal end **42** of another extension arm **36** with respect to the post **12** is less than 3.5 feet. In the lengthened position **40**, the arm span distance is greater than 7.5 feet. Each extension arm **36** is also detachable from the post **12**. The post **12** has a pair of opposed recesses **44**, each of which receives one of the extension arms **36**.

Each extension arm **36** comprises a first arm segment **48** that is coupled to the post **12** and a second arm segment **50** which is telescopically engaged to the first arm segment **48**. Each extension arm **36** also comprises a securement pin **52** which is coupled to the second arm segment **50** of an associated one of the pair of extension arms **36**. The first arm segment **36** of each extension arm **36** has an outer wall **54** and a plurality of securement holes **56** extending through the outer wall **54**. The plurality of securement holes **56** is arranged along a line extending away from the post **12** toward the distal end **42** of the associated extension arm **36**. The securement pin **52** is spring-biased to insert through one of the plurality of securement holes **56** of the associated extension arm **36**, thereby restraining axial movement of the second arm segment **50** with respect to the first arm segment **48**.

Each one of a pair of saddles **58** is coupled to the distal end **42** of an associated extension arm **36** of the pair of extension arms **36**. Each saddle **58** comprises a central member **60**, a pair of prongs **62**, and a coating **64**. The central member **60** of each said saddle **58** is coupled to and extends upwardly from the associated extension arm **36**. An upper surface **66** of the central member **60** extends transversely with respect to the associated extension arm **36** and is concavely arcuate. The upper surface **66** is configured for supporting a bar **68**. The bar **68** may be any elongated object including, for example, a curtain rod **70**. Each prong **62** is coupled to and extends upwardly from the central member **60** of an associated saddle **58** with the central member **60** extending between each prong **62**. The coating **64** is slip resistant and covers the central member **60** and each prong **62** of the associated saddle **58**. The pair of saddles **58** is configured for jointly supporting the bar **68** in a horizontal orientation **72**.

A pair of tool trays **74** is coupled to and extends downwardly from an associated one of the extension arms **36**. Each tool tray **74** comprises a panel **76**, a platform **78**, a pair of strips **80**, and a pair of hooks **82**. The platform **78** of an associated one of the pair of tool trays **74** is coupled to a top side **84** of the panel **76** of the associated tool tray **74** and has a top surface **86** which is slip resistant. The top surface **86** is configured for supporting a tool **88** and frictionally engaging the tool **88** to retain the tool **88** on the top surface **86**. Each strip **80** is coupled to one of a pair of opposing edges **90** of the panel **76** of the associated tool tray **74** and extends upwardly from the panel **76** toward the associated extension arm **36**. Each hook **82** is coupled to an upper end **92** of an associated one of the pair of strips **80** of the associated tool tray **74**. Each hook **82** extends upwardly and around the associated extension arm **36**, thereby removably hanging the associated tool tray **74** to the associated extension arm **36**. Each hook **82** is rotatable with respect to the associated strip **80** such that each hook **82** is positionable to extend downwardly from the upper end **92** of the associated strip **80**.

In use, the collapsible base **26** is positioned in the deployed position **27** and placed on the support surface **28**. The post **12** is extended or retracted to position the extension arms **36** at a height chosen by a user. The extension arms **36** are also extended or retracted to position the saddles **58** at positions chosen by the user. Then the bar **68** is positioned on the pair of saddles **58**. In this manner, the apparatus **10** may aid in the positioning of the bar **68** near a wall for attachment of the bar **68** to the wall. If the bar **68** is a curtain rod **70** for example, the apparatus **10** aids the installation of the curtain rod **70**. The tool is placed on the tool tray **74** as needed by the user. The collapsible base **26**, the post **12**, the extension arms **36**, and the tool trays **74** may be stored, for example, in a storage bag **94** when not being used.

With respect to the above description then, it is to be realized that the optimum dimensional relationships for the parts of an embodiment enabled by the disclosure, to include variations in size, materials, shape, form, function and manner of operation, assembly and use, are deemed readily apparent and obvious to one skilled in the art, and all equivalent relationships to those illustrated in the drawings and described in the specification are intended to be encompassed by an embodiment of the disclosure.

Therefore, the foregoing is considered as illustrative only of the principles of the disclosure. Further, since numerous modifications and changes will readily occur to those skilled in the art, it is not desired to limit the disclosure to the exact construction and operation shown and described, and accordingly, all suitable modifications and equivalents may be resorted to, falling within the scope of the disclosure. In this patent document, the word "comprising" is used in its non-limiting sense to mean that items following the word are included, but items not specifically mentioned are not excluded. A reference to an element by the indefinite article "a" does not exclude the possibility that more than one of the element is present, unless the context clearly requires that there be only one of the elements.

I claim:

1. A support apparatus for holding a bar, the support apparatus comprising:

a post having a top end and a bottom end, said post being extendable between a retracted position and an extended position;

a pair of extension arms, each said extension arm being coupled to said post adjacent to said top end, each said extension arm extending outwardly from said post in one of a pair of opposing directions;

a pair of saddles, each said saddle being coupled to a distal end of an associated extension arm with respect to said post, each said saddle comprising a central member being coupled to and extending upwardly from said associated extension arm, an upper surface of said central member extending transversely with respect to said extension arm, said upper surface being configured for supporting the bar, said pair of saddles being configured for jointly supporting the bar in a horizontal orientation; and

a pair of prongs, each said prong being coupled to and extending upwardly from said central member of an associated one of said pair of saddles, said central member extending between each prong said associated saddle.

2. The apparatus of claim 1, wherein said post has a height between said top end and said bottom end being less than 3.5 feet when said post is positioned in said retracted position, said height being greater than 7.5 feet when said post is positioned in said extended position.

3. The apparatus of claim 1, wherein said post comprises a plurality of rods, each said rod except for a bottommost rod of said plurality of rods being telescopically received into an adjacently lower rod of said plurality of rods.

4. The apparatus of claim 3, wherein each said rod except for said bottommost rod is frictionally retained by said adjacently lower rod with respect to said adjacently lower rod.

5. The apparatus of claim 1, further comprising a collapsible base being coupled to said bottom end of said post for supporting said post on a support surface, said collapsible base being positioned in a deployed position wherein said collapsible base extends outwardly from said post, said collapsible base being collapsible toward said post.

6. A support apparatus for holding a bar, the support apparatus comprising:

a post having a top end and a bottom end, said post being extendable between a retracted position and an extended position;

a pair of extension arms, each said extension arm being coupled to said post adjacent to said top end, each said extension arm extending outwardly from said post in one of a pair of opposing directions;

a pair of saddles, each said saddle being coupled to a distal end of an associated extension arm with respect to said post, each said saddle comprising a central member being coupled to and extending upwardly from said associated extension arm, an upper surface of said central member extending transversely with respect to said extension arm, said upper surface being configured for supporting the bar, said pair of saddles being configured for jointly supporting the bar in a horizontal orientation;

a collapsible base being coupled to said bottom end of said post for supporting said post on a support surface, said collapsible base being positioned in a deployed position wherein said collapsible base extends outwardly from said post, said collapsible base being collapsible toward said post; and

wherein said collapsible base comprises

a slider being slidably coupled to said post, said slider laterally surrounding said post,

a plurality of legs, each said leg being pivotally coupled to said slider such that each said leg is pivotable toward and away from said post, each said leg extending outwardly from said post and being angled downwardly, and

a plurality of links, each said link being pivotally coupled to said post adjacent to said bottom end and to one of said plurality of legs, wherein each said leg pivots outwardly from said post when said slider moves toward said bottom end and toward said post when said slider moves away from said bottom end.

7. The apparatus of claim 1, wherein each said extension arm being extendable away from and retractable toward said post.

8. The apparatus of claim 7, wherein said pair of extension arms is positionable in a shortened position wherein an arm span distance between a distal end of one extension arm with respect to said post and a distal end of another extension arm with respect to said post is less than 3.5 feet, said pair of extension arms being positionable in a lengthened position wherein said arm span distance is greater than 7.5 feet.

9. The apparatus of claim 7, wherein each said extension arm is detachable from said post.

10. The apparatus of claim 9, wherein said post has a pair of opposed recesses, each said opposed recess receiving one of said extension arms.

11. The apparatus of claim 7, wherein each said extension arm comprises:

a first arm segment being coupled to said post; and

a second arm segment extending telescopically into said first arm segment.

12. The apparatus of claim 11, wherein each said extension arm further comprises a securement pin being coupled to said second arm segment, said first arm segment having an outer wall and a plurality of securement holes extending through said outer wall, said plurality of securement holes being arranged along a line extending away from said post toward said distal end of said extension arm, said securement pin being spring-biased to insert through one of said plurality of securement holes, thereby restraining axial movement of said second arm segment with respect to said first arm segment.

13. The apparatus of claim 1, wherein said upper surface of said central member of each said saddle is concavely arcuate.

14. The apparatus of claim 1, wherein each saddle further comprises a coating covering said central member and each said prong of said associated saddle, said coating being slip resistant.

15. A support apparatus for holding a bar, the support apparatus comprising:

a post having a top end and a bottom end, said post being extendable between a retracted position and an extended position;

a pair of extension arms, each said extension arm being coupled to said post adjacent to said top end, each said extension arm extending outwardly from said post in one of a pair of opposing directions;

a pair of saddles, each said saddle being coupled to a distal end of an associated extension arm with respect to said post, each said saddle comprising a central member being coupled to and extending upwardly from said associated extension arm, an upper surface of said central member extending transversely with respect to said extension arm, said upper surface being configured for supporting the bar, said pair of saddles being configured for jointly supporting the bar in a horizontal orientation; and

a pair of tool trays being coupled to and extending downwardly from an associated one of said extension arms, each said tool tray comprising

a panel, and

a platform being coupled to a top side of said panel, said platform having a top surface being slip resistant, said top surface being configured for supporting a tool and frictionally engaging the tool to retain the tool on said top surface.

16. The apparatus of claim 15, wherein each said tool tray further comprises:
a pair of strips, each said strip being coupled to one of a pair of opposing edges of said panel, each said strip extending upwardly from said panel toward said associated extension arm; and
a pair of hooks, each said hook being coupled to an upper end of an associated one of said pair of strips, each said hook extending upwardly and around said associated extension arm, thereby removably hanging said tool tray to said associated extension arm.

17. The apparatus of claim 16, wherein each said hook is rotatable with respect to said associated strip, each said hook being positionable to extend downwardly from said upper end of said associated strip.

18. The apparatus of claim 1, further comprising:
said post being extendable between a retracted position and an extended position; said post having a height between said top end and said bottom end being less than 3.5 feet when said post is positioned in said retracted position, said height being greater than 7.5 feet when said post is positioned in said extended position, said post comprising a plurality of rods, each said rod except for a bottommost rod of said plurality of rods being telescopically received into an adjacently lower rod of said plurality of rods, each said rod except for said bottommost rod being frictionally retained by said adjacently lower rod with respect to said adjacently lower rod;
a collapsible base being coupled to said bottom end of said post for supporting said post on a support surface, said collapsible base being positioned in a deployed position wherein said collapsible base extends outwardly from said post, said collapsible base being collapsible toward said post, said collapsible base comprising:
a slider being slidably coupled to said post, said slider laterally surrounding said post;
a plurality of legs, each said leg being pivotally coupled to said slider such that each said leg is pivotable toward and away from said post, each said leg extending outwardly from said post and being angled downwardly; and
a plurality of links, each said link being pivotally coupled to said post adjacent to said bottom end and to one of said plurality of legs wherein each said leg pivots outwardly away from said post when said slider moves toward said bottom end and toward said post when said slider moves away from said bottom end;
each said extension arm being extendable away from and retractable toward said post, said pair of extension arms being positionable in a shortened position wherein an arm span distance between a distal end of one extension arm with respect to said post and a distal end of another extension arm with respect to said post is less than 3.5 feet, said pair of extension arms being positionable in a lengthened position wherein said arm span distance is greater than 7.5 feet, each said extension arm being detachable from said post, said post having a pair of opposed recesses, each said opposed recess receiving one of said extension arms, each said extension arm comprising:
a first arm segment being coupled to said post;
a second arm segment extending telescopically into said first arm segment; and
a securement pin being coupled to said second arm segment, said first arm segment having an outer wall and a plurality of securement holes extending through said outer wall, said plurality of securement holes being arranged along a line extending away from said post toward said distal end of said extension arm, said securement pin being spring-biased to insert through one of said plurality of securement holes, thereby restraining axial movement of said second arm segment with respect to said first arm segment;
each said saddle comprising:
a coating covering said central member and each said prong, said coating being slip resistant;
a pair of tool trays being coupled to and extending downwardly from an associated one of said extension arms, each said tool tray comprising:
a panel;
a platform being coupled to a top side of said panel, said platform having a top surface being slip resistant, said top surface being configured for supporting a tool and frictionally engaging the tool to retain the tool on said top surface;
a pair of strips, each said strip being coupled to one of a pair of opposing edges of said panel, each said strip extending upwardly from said panel toward said associated extension arm; and
a pair of hooks, each said hook being coupled to an upper end of an associated one of said pair of strips, each said hook extending upwardly and around said associated extension arm, thereby removably hanging said tool tray to said associated extension arm, each said hook being rotatable with respect to said associated strip, each said hook being positionable to extend downwardly from said upper end of said associated strip.

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